

Traffic Congestion at Roxbury Crossing

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Abstract

Traffic congestion in Boston significantly impacts daily commutes, air quality, and urban mobility. This study analyzes traffic delays and levels of service (LOS) at the Roxbury Crossing intersection using simulation models. Signal timing is also optimized based on simulation inputs and outputs to reduce average delays. The analysis was performed using Synchro and Vistro, incorporating traffic volume data from 2018 and signal timing plans from both 2018 and 2025. Since the available traffic counts may not reflect current conditions, a sensitivity analysis was conducted using three different volume scenarios. This approach evaluates how changes in traffic volumes across different lanes affect overall intersection performance.

1 Introduction

1.1 Background and Motivation

The intersection of Tremont Street, Columbus Avenue, and Malcolm X Boulevard in Roxbury Crossing, Boston, is a major transit hub that facilitates the movement of daily commuters, residents, and commercial traffic [1]. This intersection experiences substantial traffic volumes, particularly during peak hours, leading to congestion and delays for private vehicles and transit buses.

A primary concern at this location is the inconsistent delay experienced by buses. Due to the timing of traffic signal cycles, buses arriving at different times encounter varying levels of delay. While some may pass through with minimal wait, others can be delayed for over 100 seconds, especially during periods of heightened congestion. These inconsistencies suggest inefficiencies in the intersection’s operations, adversely affecting both transit reliability and overall traffic flow.

This study evaluates the intersection’s performance by analyzing 2018 traffic conditions, along with signal timing plans from both 2018 and 2025. Through simulation modeling and sensitivity analysis, we explore potential optimizations to signal timing and phasing aimed at reducing average delays and improving the overall level of service (LOS).

Our motivation stems from both a recognized gap in the literature and a personal connection to the issue—as residents of Boston, we regularly experience the impacts of this intersection’s performance firsthand and are committed to contributing toward its improvement.

1.2 Objectives of the Study

The primary objective of this study is to assess the performance of the Roxbury Crossing intersection under existing traffic conditions and explore potential operational improvements. Using 2018 traffic volume data and signal timing plans from both 2018 and 2025, the study focuses on analyzing vehicle delays and levels of service (LOS).

Simulation tools such as Synchro and Vistro are used to model traffic operations and evaluate various signal timing scenarios. Additionally, a sensitivity analysis is conducted

by adjusting traffic volumes across different lanes to examine how performance metrics respond to changes in demand.

2 Methodology

2.1 Data Collection

To conduct a comprehensive analysis, the study will employ a combination of data collection and simulation-based evaluation techniques. The methodology that is included in this research is as follows:

- **Traffic Volume Data Collection:** Peak-hour traffic volume data will be obtained from the Boston government department. The dataset includes turning movement counts for all approaches at the intersection.
- **Field Observations of Traffic Signals:** On-site data collection will involve recording traffic signal cycle lengths, phase sequences, and vehicle movements at the intersection.
- **Synchro Analysis:** The Synchro software will be used to perform capacity analysis, determining the level of service and delay for each movement at the intersection. This analysis will also allow for the simulation of signal timing modifications to evaluate potential improvements in traffic efficiency.
- **Vistro:** Vistro software will be used to perform capacity analysis, determining the level of service and delay for each movement at the intersection. This analysis will also allow for the simulation of signal timing modifications to evaluate potential improvements in traffic efficiency.

3 Traffic Patterns and Study Scope

3.1 Morning Peak Hour Focus

Boston experiences two main congestion peaks:

- **Morning (7:00 - 9:30 AM):** Inbound traffic towards business districts.

- **Evening (4:30 - 7:00 PM):** Outbound congestion on major highways.

For this study, the analysis focuses on a one-hour window between 7:00 and 8:00 AM to capture morning peak conditions.

3.2 Considerations

While this study aims to provide actionable insights for improving intersection performance, several assumptions and limitations must be noted.

This analysis is conducted as a terminating simulation, focusing on a defined one-hour morning peak period between 7:00 and 8:00 AM. This approach allows for a detailed snapshot of traffic operations during a critical time window but does not capture conditions throughout the full day or across different seasons.

Although the primary emphasis is on vehicular traffic, pedestrian movements were accounted for in the signal phasing design used in the simulation models. However, the study does not include a detailed evaluation of pedestrian or bicycle flow, which may influence intersection efficiency and safety.

Furthermore, the models do not incorporate external variables such as weather, special events, or road construction, all of which can significantly impact traffic dynamics. Despite these constraints, the use of real-world volume data, scenario testing, and advanced simulation tools supports a robust evaluation of the intersection and offers valuable insights into signal timing optimization.

4 Case Study: Traffic Congestion in Roxbury Crossing

4.1 Site Description



Figure 1: Roxbury Crossing Intersection [2]

The Roxbury Crossing intersection is surrounded by several schools, colleges, and a bus station:

- Northeastern University
- Roxbury Community College
- Fenway High School
- Boston Adult High School
- James P. Middle School

- Nubian (Bus Station)

On top of that, there are numerous bus lines that pass through the intersection every day, along with many bus stops, making the area a hotspot for traffic congestion and a high percentage of heavy vehicles (mainly buses, see Figure 2).

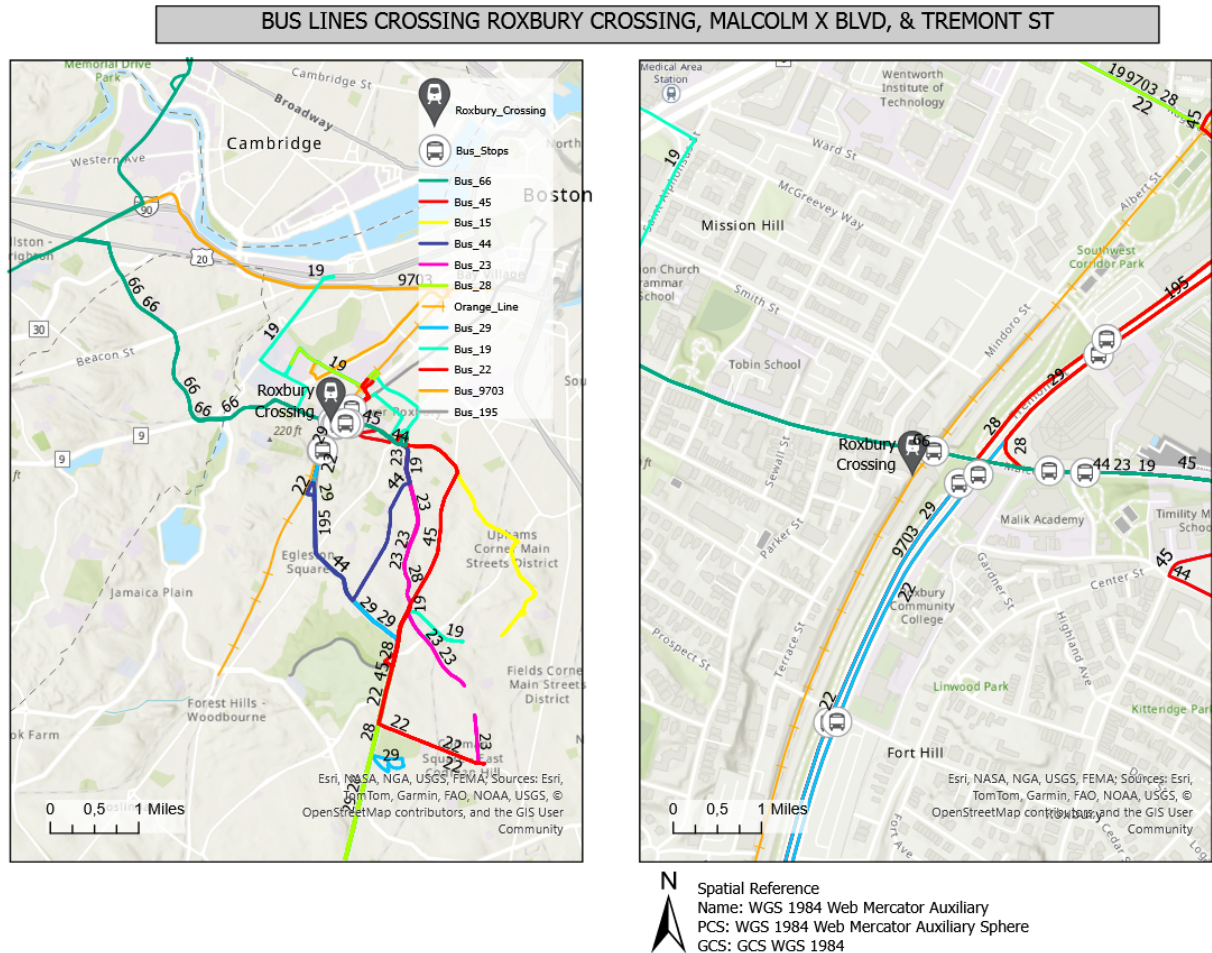


Figure 2: Bus Lines and Bus stops (own creation)

We also created a sketch of the intersection (see Figure 3) to better understand the geometry and lane directions. This sketch was used to define the lane configuration within the simulation environment.

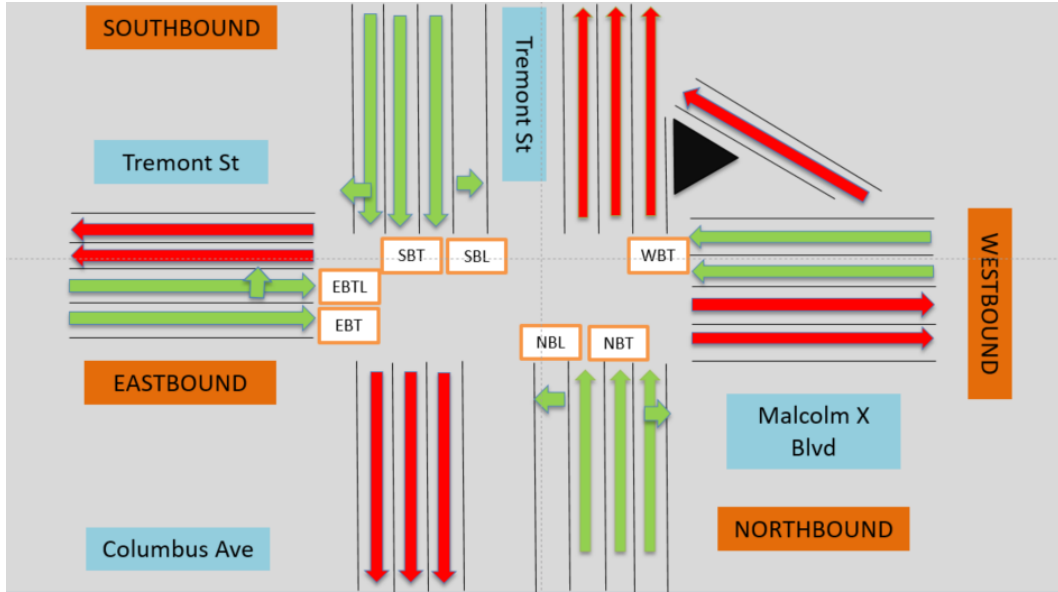


Figure 3: Sketch of the Roxbury Crossing Intersection and its Lanes (own creation)

4.2 Volume Counts and Signal Timing

The traffic counts of vehicles from Precision Data Industries for 2018, between 7 and 8 AM, were used as input for our analysis [3]. Additionally, we used the signal timing data from 2018 provided by the Massachusetts Department of Transportation [4] as a second input for our initial analysis using Synchro.

Class:

Cars and Heavy Vehicles (Combined)																							
Tremont Street						Malcolm X Boulevard						Columbus Avenue						Tremont Street					
from North						from East						from South						from West					
Right	Thru	Left	U-Turn	Total		Right	Thru	Left	U-Turn	Total		Right	Thru	Left	U-Turn	Total		Right	Thru	Left	U-Turn	Total	
7:00 AM	28	78	22	1	129	124	74	13	0	211	32	268	40	0	340	21	64	30	0	115	795		
7:15 AM	42	128	19	3	192	116	61	16	0	193	15	333	41	1	390	27	48	21	0	96	871		
7:30 AM	51	89	22	5	167	94	80	5	0	179	13	286	34	0	333	19	65	48	0	132	811		
7:45 AM	31	111	22	0	164	144	81	5	0	230	7	328	49	0	384	30	51	33	0	114	892		
Total	152	406	85	9	652	478	296	39	0	813	67	1215	164	1	1447	97	228	132	0	457	3369		

Figure 4: Traffic Volume Counts [3]



Figure 5: Signal Timing Phases 2018 [4]

On the other hand, since our analysis using Synchro did not accurately represent reality due to unrealistic delay times (too high delays, see Figure 7), we decided to collect the new signal timing for 2025 in person (see Figure 6).

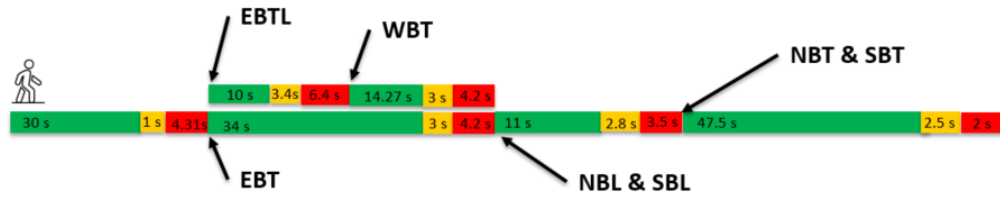


Figure 6: Signal Timing Phases collected 2025 (own creation)

4.3 Simulation Base Models and Results

For the initial simulation using Synchro with the 2018 volume data and signal timing, we observed an average delay of 300.9 seconds at the intersection and a level of service (LOS) of F. This indicated an unrealistic delay and very poor performance.



Figure 7: Roxbury Crossing Intersection - Synchro

Given these results, we decided to pivot from Synchro to Vistro for our analysis, as it was becoming increasingly difficult to perform accurate simulations with Synchro. Using the same 2018 volume data and the updated 2025 signal timing, we obtained a closer representation of reality. However, there was still considerable bias, as we observed in

the field that the Northbound and Southbound left lanes experienced the highest levels of congestion.

For this reason, we concluded that the 2018 volume data was not representative of 2025 conditions. Acquiring new volume data would require fieldwork, including the installation of cameras, loop detectors, or other technologies/devices to count traffic. This process would be both time-consuming and costly. Therefore, we decided to perform a sensitivity analysis with three different scenarios to better understand the performance of the intersection under varying conditions. The next section provides details on both the signal optimization and the sensitivity analysis.

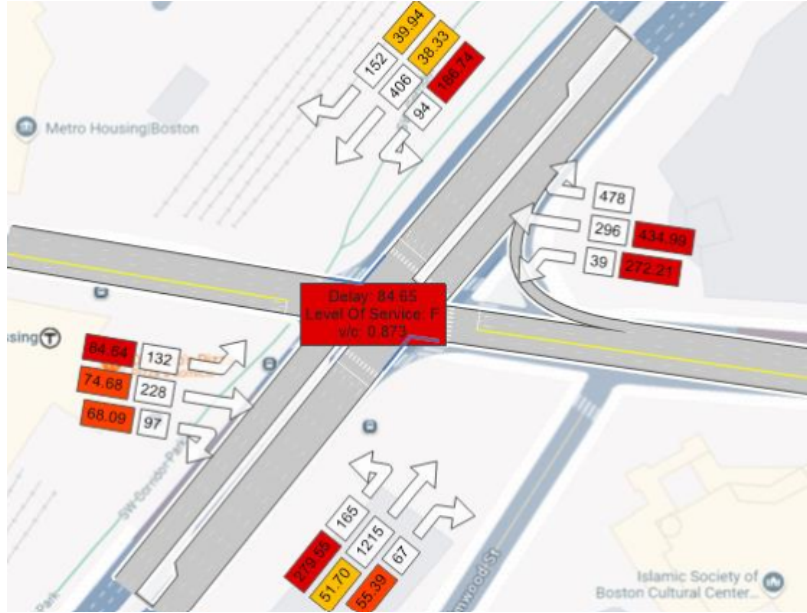


Figure 8: Roxbury Crossing Intersection - Vistro

5 Signal Optimization

Signal optimization sought to improve intersection performance by lowering total delays and raising the Level of Service (LOS) at Roxbury Crossing. At first, the assessment concentrated on the separate pedestrian phase, which lasted 34 seconds and greatly affected car delay. Vehicle movements were completely stopped during this exclusive pedestrian phase, greatly adding to the total delay of the intersection.

Discrepancies appeared when I gathered fresh on-field signal timing data for this study in comparison to the prior 2018 version run by the city. Interestingly, the city kept the

only pedestrian phase even with these modifications. A thorough study was therefore conducted to investigate the justification for keeping this phase. Roughly 100-105 feet, the longest pedestrian crosswalk at the intersection. Guidelines in the Manual on Uniform Traffic Control Devices (MUTCD) say that pedestrian walking speeds usually vary from 3.0 ft/s for elderly pedestrians to 3.5 ft/s for typical pedestrians. At least 30 seconds of pedestrian walk phase length was therefore considered required, supporting the ongoing use of an exclusive phase for pedestrians.

Briefly examined were alternative scenarios, including matching pedestrian green phases with parallel vehicle traffic or using leading pedestrian intervals. Given that the city's updated signal plan had already turned down such choices, maybe for safety reasons, these were not followed much. Another factor was the possible danger from vehicle turning speeds, especially at a major intersection like Roxbury Crossing, where several approaches have 3-4 lanes each, so increasing the danger of pedestrian crossings.

Signal cycle lengths were tested by raising the baseline cycle length from 150 seconds to 160 seconds and lowering it to 140 seconds. An ideal allocation was found by means of repeated testing (Figure 9). Particularly, the Westbound left-through-right (LTR) movement got 9 seconds more in green time since this was the only method, all turning movements shared a single green phase. The Eastbound left turn phase, on the other hand, was cut by 2 seconds without affecting general intersection performance. Since Northbound-Southbound (NB-SB) directions have the most traffic during the AM peak, their through-right movements already have a significant green time of 52 seconds. Thus, more accessible seconds were redistributed to improve the Northbound left (NBL) and Southbound left (SBL) movements, increasing their split from 17 seconds to 26 seconds.



Figure 9: Optimized Signal Timing Plan

The results of the optimization showed significant gains (Figure 10). From an initial 84.65 seconds to 75.45 seconds, the total intersection delay dropped, producing a notable

savings of 9.2 seconds per cycle, or a 10.8% drop. The LOS also changed from grade F to grade E, suggesting improved operational performance. Furthermore, the volume-to-capacity (v/c) ratio notably changed from 0.873 to 0.679, implying less congestion and more efficient use of the intersection's capacity.

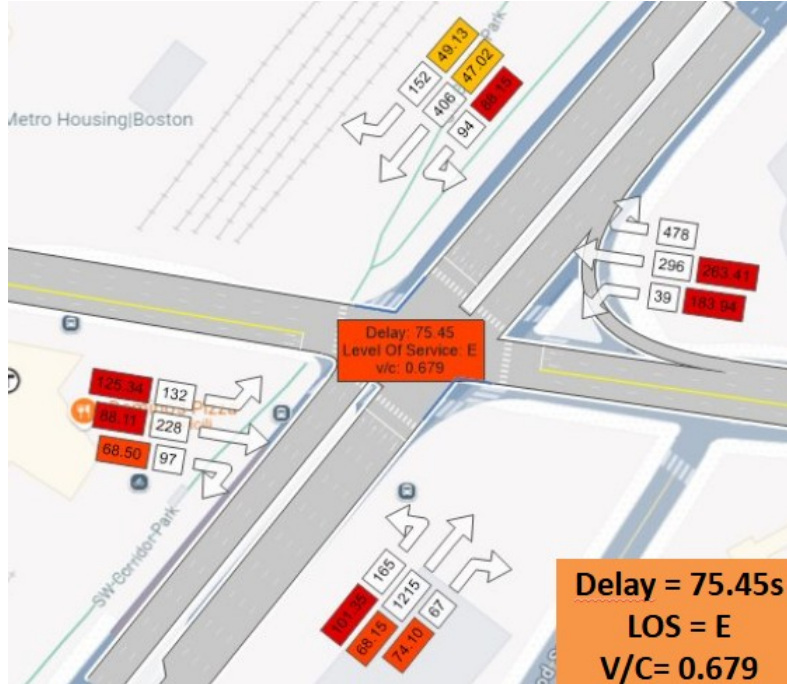


Figure 10: Optimized Version for Intersection Layout

6 Sensitivity Analysis

After the signal timing plan was successfully optimized, a sensitivity study was done to evaluate how changes in traffic volume would affect the intersection delays and general performance at Roxbury Crossing. This study sought to shed light on the strength of the optimized signal timings under different demand scenarios.

With a v/c ratio of 0.679 after optimization, the intersection is not under very high-stress demand, which usually happens when v/c ratios approach or exceed 0.95. The volume data used for the study came from 2018, thus, major changes might have taken place since then, affected by nearby road improvements, population increase, or changing traffic patterns. Therefore, scenarios were created using volume multipliers to mirror possible demand changes.

Simulating a situation of lower demand, the first sensitivity scenario called for a 0.9

multiplier reduction of all turning movement volumes (Figure 11). The junction saw a notable drop in delay to 66.08 seconds under this condition. This outcome indicates that even a small drop in traffic levels greatly improves intersection performance, indicating that small drops in traffic demand could significantly reduce congestion.

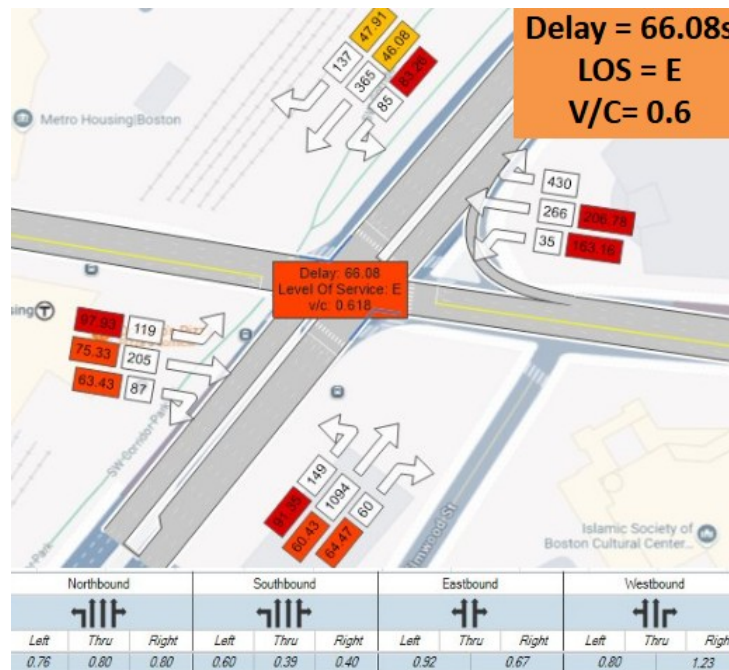


Figure 11: Optimized Version (0.9x)

A higher demand situation was represented in the second scenario using a volume multiplier of 1.3 (Figure 12). Delays climbed significantly to 131.42 seconds under this higher load. Even though at this increased demand, the v/c ratio was 0.85, suggesting the intersection had not yet reached critical capacity limits. This result suggests that although under greater demand circumstances, intersection performance would suffer greatly, it still has the operational function, albeit at lower efficiency and comfort.

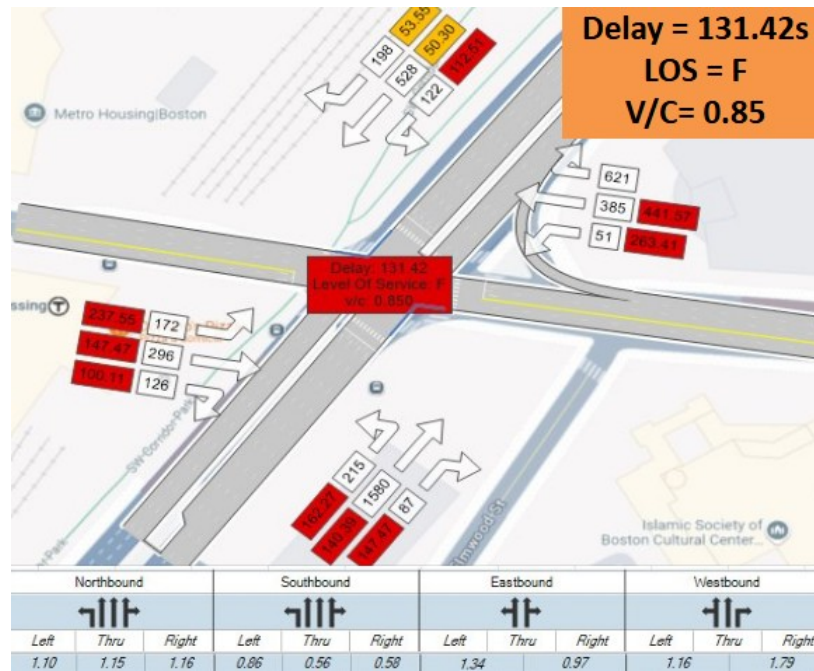


Figure 12: Optimized Version (1.3x)

A third scenario examined a hybrid situation, using a lower 0.9 multiplier generally but raising volumes particularly on the major Northbound and Southbound approaches with a multiplier of 1.5 (Figure 13). Intersection performance was greatly affected by this focused rise, which produced a total delay of 120.06 seconds and a v/c ratio of 0.785. The result emphasizes the sensitivity of the intersection's performance to volume changes on these main axes. It emphasizes the need to control traffic flow, especially on the vital Northbound and Southbound directions, since higher volumes disproportionately raise congestion and delays.

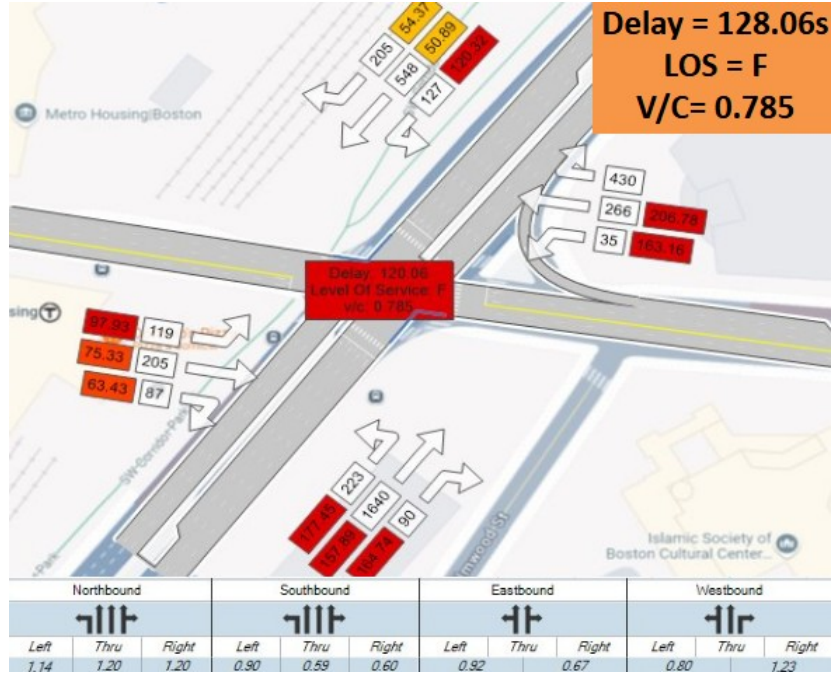


Figure 13: Optimized Version ($1.5 \times$ NB/SB and $0.9 \times$ EB/WB)

The sensitivity study highlights that while the optimized intersection can effectively handle present demand levels, it is sensitive to future traffic volume increases, especially on important directional flows. These findings therefore imply that to maintain acceptable levels of service under expected traffic growth, continuous monitoring and proactive actions like adaptive signal control or infrastructure improvements could be required.

7 Interpretation

The findings from both the optimization and sensitivity studies provide significant new perspectives on traffic control at the Roxbury Crossing junction. The optimization clearly showed that changing signal timings reduces delays and improves the Level of Service (LOS), therefore greatly improving intersection performance under current conditions. Reflecting a LOS grade of F and a v/c ratio of 0.873, the intersection initially ran with a significant delay of 84.65 seconds per cycle. After optimization, delays dropped by 10.8% to 75.45 seconds per cycle, so raising the LOS from grade F to grade E and lowering the v/c ratio to 0.679. These results draw attention to the direct advantages of putting the improved signal plan into effect.

Moreover, the sensitivity study emphasized the intersection's reaction to volume fluctu-

ations, therefore stressing important thresholds and directions especially susceptible to congestion. The notable differences in intersection delay under changed traffic demand highlight the need for flexible, adaptive management strategies. Particularly remarkable was the effect on performance when Northbound and Southbound major directional flows saw rising volume, implying that specific management policies emphasizing these main approaches could produce considerable operational advantages.

The results show generally the possibility of better operational efficiency at Roxbury Crossing by means of strategic traffic signal management, and they strongly support proactive, adaptive planning techniques. Though the intersection would gain greatly from continuous evaluation and adaptive techniques to control changing traffic dynamics, the present infrastructure and signal settings of the intersection can handle current conditions sufficiently.

8 Future Work and Limitations

This study has to be acknowledged as having several limitations. First off, the main drawback is the application of 2018 antiquated traffic volume data, which could cause errors in representing present circumstances. Urban development, infrastructure projects, seasonal changes, and local events, all not specifically addressed in this study, can greatly alter traffic dynamics. Human mistakes in data collecting, analysis, or input can also cause variations that compromise the precision and repeatability of findings. The analysis's own reproducibility is difficult since consistent findings could not be certain when using different adaptive signal control techniques or when assessing the intersection performance at different times or under different seasonal and event-specific circumstances.

To more reliably reflect present traffic conditions and more accurately forecast future scenarios, future studies should give the gathering of current and correct traffic volume data top priority. Examining lane allocation or configuration changes as part of a lane redesign could help to improve traffic flow and safety. Moreover, thorough geometric studies of the intersection should be carried out to find design improvement possibilities, therefore lowering congestion and improving pedestrian safety. Finally, looking at relationships with adjacent intersections could offer insightful analysis of more general network impacts, therefore guaranteeing harmonized signal timing plans that enhance

regional traffic flow and lower system delays.

9 Conclusions

Focusing on maximizing signal timing to enhance operational efficiency, this paper offered a thorough assessment of the Roxbury Crossing junction. Important results indicate that using the suggested signal timing changes caused a 10.8% decrease in delay and raised the intersection's Level of Service from grade F to E. These findings verify that, even under complicated traffic patterns, focused signal timing improvements can greatly reduce congestion and enhance performance.

Apart from the obvious advantages, this study emphasizes the need for ongoing monitoring and adaptive signal control. Particularly along the Northbound and Southbound approaches, the sensitivity study showed how rapidly performance could decline under fluctuating demand. These results support including data-driven updates and adaptive signal control techniques to guarantee the intersection keeps operating effectively as conditions change.

References

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